

Invention*ID*

Automated Invention Extraction and Prior-Art Analysis from Technical Documents

TEAM

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MENTORS

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→ github.com/VasistP/InventionID

A reviewer starts with *a paper*, not a patent search.



FIRST-PASS REVIEW

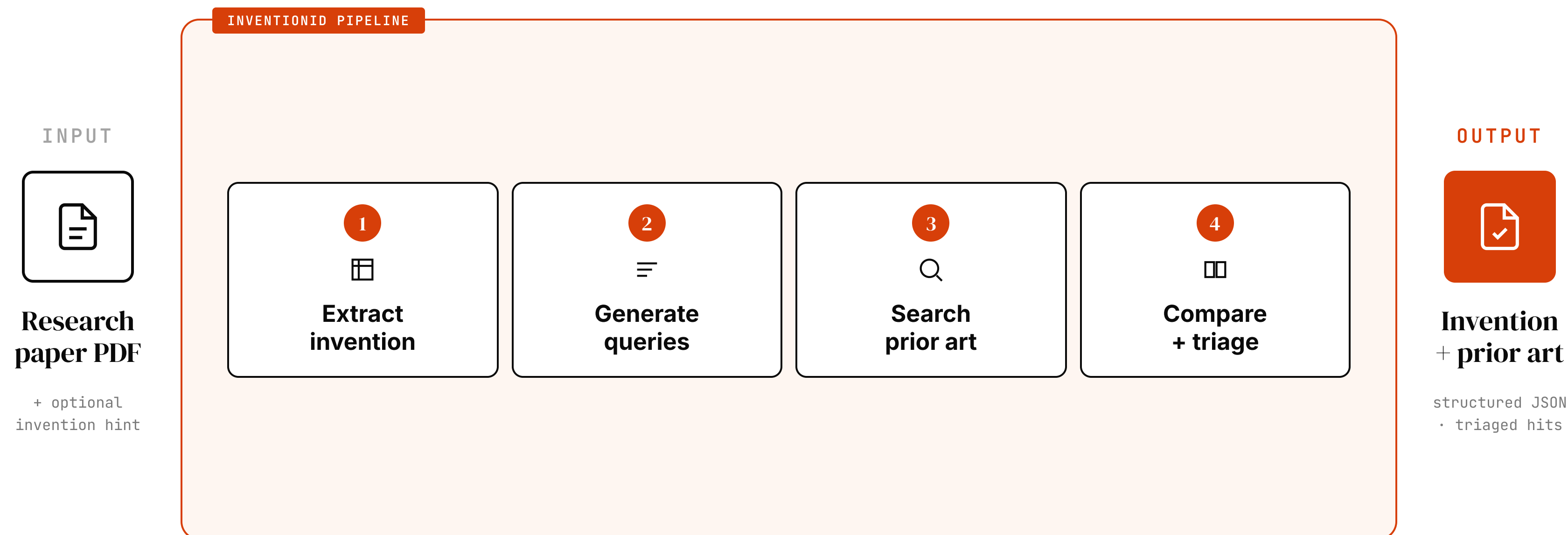
1 Is this a patentable invention?

ANSWERED BY EXTRACTION

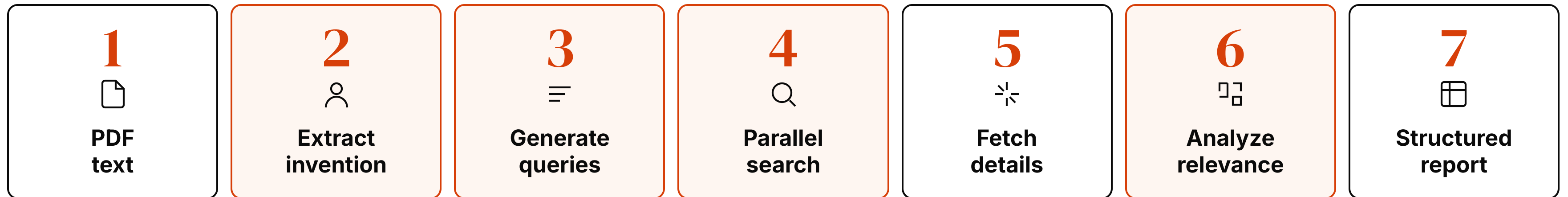
2 What related prior art exists?

ANSWERED BY SEARCH + TRIAGE

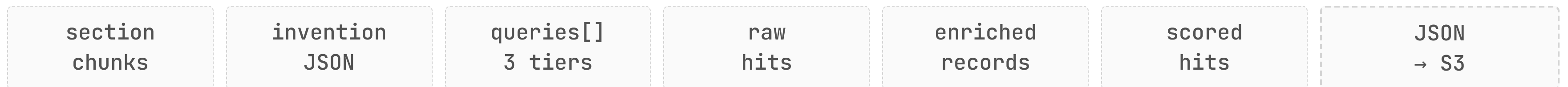
Paper goes in. *Review-ready* prior art comes out.



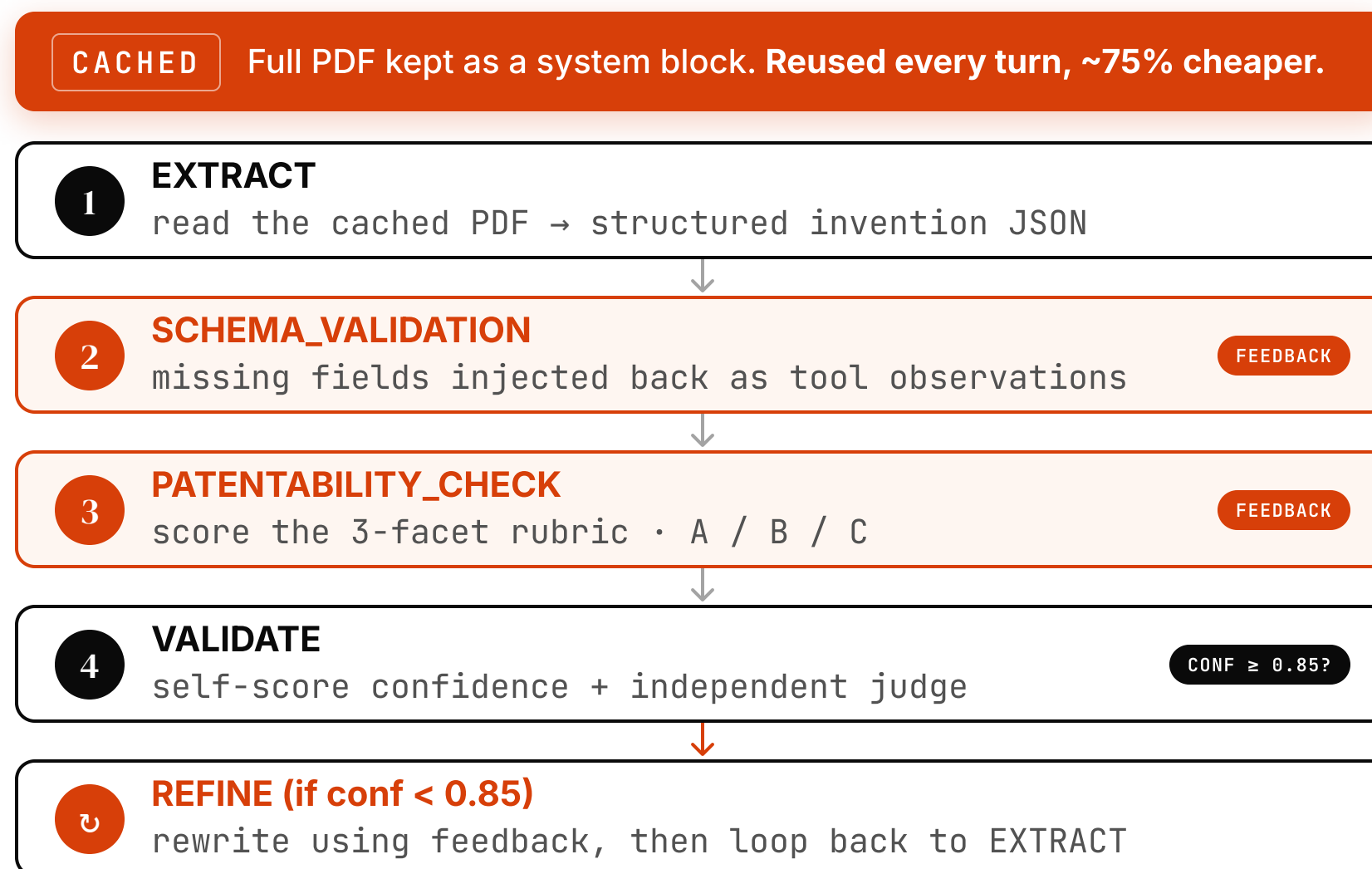
From PDF to *structured report*.



EACH STAGE PRODUCES AN INTERMEDIATE ARTIFACT • INSPECTABLE • ADJUSTABLE



A *ReAct loop* with cached context.



1 PROMPT CACHING

Reuse the full PDF
across every turn.

Claude Sonnet

ephemeral cache

~75% cost cut

2 TOOL OBSERVATIONS

Failures become turns,
not retries.

schema_validator

patentability_classifier

3 MAX 4 ITERATIONS

Bounded loop, exits
on confidence ≥ 0.85 .

Drop a PDF. Watch the *pipeline run*.

1 Upload paper
drag PDF · optional invention hint

InventionID · New Analysis

alysis < Dashboard **Analysis**

New Analysis

analyses yet

Patent Prior Art Analysis
Upload a research paper or patent document (PDF) to analyze for prior art and patentability.


Drop a PDF here or click to browse

What do you think your invention is? (optional)
Briefly describe the invention in your own words...

Sign out

2 Live pipeline progress
7 stages · transparent + auditable

InventionID · Processing

Stage 1 of 7: Extracting text from PDF

Extracting text from PDF

Extracting invention details

Generating search queries

Searching patents

Fetching patent details

Analyzing patents

Generating final report

The agent's *actual output*.

Dashboard
Analysis

✓ Analysis Complete

Microbial Electrosynthesis of Single Cell Protein Using Methanococcus maripaludis

Biotechnology - Microbial Electrosynthesis and Bioprocessing
Process

A system coupling fast-growing methanogen *Methanococcus maripaludis* with microbial electrolysis cells (MECs) to produce single cell protein (SCP) and methane. The MECs generate hydrogen in situ from organic waste (acetate) via electroactive bacteria on anodes, which *M. maripaludis* consumes for rapid growth and protein synthesis. The process achieves a doubling time of 11.2 hours, specific growth rate of 0.062 1/h, and SCP production rate of 13.7 mg/L/h with over 60% protein content in dried biomass.

Key Technical Features

- Single chamber MEC with stacked electrode assembly: two MoP-coated cathodes and two carbon cloth anodes with 12.5 cm² projected surface area each
- In situ hydrogen generation at 2.5-3.8 L/L/D via electrochemical process at 1.0 V applied voltage, maintaining hydrogen partial pressure below 5 kPa

A

Title + description

domain · summary · key claims ·
written in plain English

B

Key technical features

structured bullets · auditable
by reviewer

C

Patentability tag

3-facet rubric · 6/6 in this
case

D

Hit + triage counts

20 patents · 42 papers · triage
rollup

Every stage has a *cheaper, simpler* alternative.

EXTRACTION STRATEGY



ReAct loop
validate → refine

vs



Single-shot
prompt

One pass misses fields and misjudges patentability. The loop **self-corrects to 85% confidence.**

Every stage has a *cheaper, simpler* alternative.

QUALITY CHECK



Independent
judge (Llama)

vs



Self-assessment
only

A model grading its own work is biased. A **different model family catches blind spots.**

Every stage has a *cheaper, simpler* alternative.

DOCUMENT CONTEXT



Prompt
caching

vs



Re-send paper
each turn

A 40-page paper is ~150K tokens. Caching makes iterations **~90% cheaper**.

Every stage has a *cheaper, simpler* alternative.

PATENT SEARCH



10 parallel
tiered queries

vs



One keyword
search

Paper and patent vocabularies differ. Tiered queries **bridge the gap, in parallel.**

A 3-facet rubric, scored 0 to 6.



Invention Type

Method, system, product, or composition?



Patent Eligibility

Concrete mechanism, or abstract idea?



Practical Application

Measurable gain over a baseline?

TOTAL SCORE → VERDICT

0–2

Scientific Discovery

primarily theoretical

3–4

Borderline Case

partial mechanism or data

5–6

Potential Invention

implemented + measurable

HARD GUARDRAIL

If **A = 0** or **B = 0**, classify as Scientific Discovery and skip the total.

Turn paper words into *broad patent queries*.

1

Take the paper's words

"microbial protein" · "bioelectrochemical synthesis"



2

Expand with synonyms

+ SCP biomass MEC electrofermentation



3

Generate 10+ queries

specific → adjacent → broader
each tagged with hit count



RUN IN PARALLEL

Google Patents + Scholar

5 workers each

InventionID · Search Query Calibration		
Search Query Calibration (10 queries)		
1	Methanococcus maripaludis AND ("microbial electrosynthesis" OR "bioelectrochemical synthesis" OR electrofermentation) AND ("single cell protein" OR SCP OR "microbial protein")	1 · Too specific
2	Methanococcus maripaludis AND ("microbial electrolysis" OR MEC OR "bioelectrochemical system") AND hydrogen	16 · Very specific
3	methanogen AND ("microbial electrolysis" OR MEC OR "bioelectrochemical system") AND ("protein production" OR "biomass production" OR "cell growth")	38 · Very specific
4	("single cell protein" OR SCP OR "microbial protein") AND ("in situ" OR "in-situ") AND ("hydrogen generation" OR "hydrogen production" OR "H2 generation")	40 · Very specific
5	("microbial electrosynthesis" OR "bioelectrochemical synthesis" OR electrofermentation OR "microbial electrolysis") AND acetate AND ("protein production" OR "biomass synthesis")	15 · Very specific
6	hydrogenotrophic AND methanogen AND ("electrochemical hydrogen" OR "cathodic hydrogen" OR "electrolytic hydrogen") AND ("protein synthesis" OR "biomass growth")	1 · Too specific
7	("single cell protein" OR SCP OR "microbial protein") AND chemoautotrophic AND ("waste conversion" OR "waste treatment" OR "organic waste")	21 · Very specific
8	("microbial electrolysis" OR MEC OR "bioelectrochemical system") AND cathode AND ("hydrogen consumption" OR "hydrogen utilization") AND (methanogen OR archaeon)	9 · Too specific

Patents *and* papers · three buckets each.

01 · POTENTIALLY BLOCKING

02 · RELEVANT

03 · RELATED

PATENT RESULTS

Dashboard

Analysis

Potentially Blocking (1)

Microbial electrosynthesis of single cell protein

US20240140987A1

claim 16 , wherein the electrolyte has a retention time of 4 hours to 48 hours in the MEC. 18 . The method of claim 1 , further comprising separating the SCP from the electrolyte. 19 . The method of claim 18 , wherein: (i) separating the SCP from the electrolyte comprises electrocoagulation, adding a ...

Relevance 95%

Detailed Analysis

Filed: 2023-10-24 Assignee: Oregon State University

View Patent

Relevant (6)

Enhanced microbial electrosynthesis by using co-cultures

US20170166883A1

12 . The microbial electrosynthesis system of claim 1 , wherein the first microorganism is a hydrogen producing microorganism or a formate producing organism. 13 . The microbial electrosynthesis system of claim 12 , wherein the hydrogen producing microorganism has a hydrogenase enzyme. 14 . The ...

Relevance 75%

Detailed Analysis

ACADEMIC PAPERS

Dashboard

Paper results

Microbial conversion of co2 and other c1 substrates to vegan nutrients, ...

US20250066715A1

53 . The method according to claim 52 , wherein electrolysis of water for the production of hydrogen is performed using one or more of: Proton Exchange Membranes (PEM); liquid electrolytes such as KOH; alkaline electrolysis; Solid Polymer Electrolyte electrolysis; high-pressure electrolysis; and high ...

Relevance 35%

Detailed Analysis

Filed: 2024-10-25 Assignee: Kiverdi, Inc.

View Patent

Other (5)

Academic Papers (42)

Potentially Blocking (1)

Microbial electrosynthesis of single cell protein and methane by coupling fast-growing Methanococcus maripaludis with microbial electrolysis cells

L Wang, H Liu - Bioresource Technology, 2024 - Elsevier

Relevance 100%

Detailed Analysis

View Paper

What *InventionID* delivers.

1

Hours become minutes

A first-pass review that once took hours of manual searching now runs in minutes.

2

Answers the core questions

Surfaces whether there is a patentable invention, and what prior art already exists.

3

Transparent by design

Every stage, from extraction to queries to results, is visible and can be adjusted.

4

Supports, never replaces

Accelerates the reviewer's triage and leaves the final judgment to the human.

Where *InventionID* goes from here.

1

Secure deployment

HTTPS hosting with authentication for real reviewer use.

2

Broader corpora & export

Add more patent sources and a polished PDF report export.

Demo.

■ THANK YOU ■

Questions?

InventionID: a paper-first workflow for invention extraction & prior-art analysis.



Oregon State University
College of Engineering

TEAM

Pranav Vasist · Pranav Mane · Prayoga

REPO github.com/VasistP/InventionID